

Tracking Individual Polarization and Aggression: A Longitudinal Study of Reddit Super-Users

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1 Abstract

The pace of growth of online forums has led to online discourse being a high frequency, decentralized market with the power to mobilize collective action on a grand scale. This study draws inspiration from the fact that social media now highly effects the public opinion, and analyze the political posts from Reddit to determine the user aspects and the structural tendencies of the posts.

Through a comprehensive dataset of 13,029,405 sub engagements on the political subreddits through the 2024 Election Cycle of the United States (September 2023 - September 2025), a machine learning process will be deployed to identify 309,956 messages of 2,495 users with 500 or more actions. Then, the "Hyper-Core" user groups were identified as those who assess the systemic nature of political radicalization and aggression. To evaluate the aggressiveness of messages on the platform, this research will establish continuous measures of political polarization through Sentence-BERT and weak supervision.

The findings indicate a strong causal inversion between the different tiers of users. While for regular users aggression is a gateway emotion to partisan switching, for high-risk agitators, ideological inflexibility is the prime motivator for strategic aggression. Moreover, this research shows that high-risk behavior is predictable. The model from this study gained an accuracy of 62% on agitator classification, which was attained by an XGBoost model relying solely on metadata-driven variables, with average net engagement (clout) being the best predictor of toxic content.

These findings are conclusive about the "Ratchet Effect", which the design parameters of the Reddit platform act as a inducement for increased community toxicity by promoting aggressive partisan performative action as a means to gain overall community influence. This research aims to fill a significant conceptual gap regarding early warning systems for monitoring social radicalization incidents within online web communities.

2 Introduction

In the new paradigm of the online world, politics has been revolutionized into a high-frequency decentralized market with strong partisan identification and rhetorical dynamics. This online communities now became one of the most important influence to the public opinion, while many politicians utilize this environment. This has increased the power of internet communities from mere online discussions into high-powered engines that have the capability of bringing massive collective action and fundamental behavioral changes.

I personally studied on the "meme stock" event of 2021, with special emphasis on the GameStop(GME) event, which presented a paradigm-shifting case on the potential power of the online community Reddit. Now many retail investors on platforms like this generate massive collective action outside the age-old dynamics of

financial market efficiency. By this study, I realized the importance of online community towards public opinion, and how to address which users create the biggest influence. Since politics are one of the biggest subject as well as a direct index that can be projected to reality.

The basis for this research uses an exhaustive interaction set extracted for the two main ideological convention on the Reddit platform via Kaggle, r/Democrats and r/Republicans, as well as other political subreddits. The main analysis was exactly targeted to cover the period of September 1, 2023, to September 30, 2025, which covers the 2024 election period as well as the pre-election era and the post-election era.

In the first stage of the data extraction, an enormous set of 13,029,405 interaction events was obtained, of which 4,977,517 were drawn from the Democratic categories and 8,051,888 from the Republicans. One of the most important aspects of this analysis was to determine the "Hyper-Core," the users who strongly interfere in the creation of public opinion in this online platform.

The main intention of this research is to explore the measure in which the structural motivation on Reddit influences market-like irregularities in the field of politics. The proposed project will be able to satisfy several specific objectives, including examining the directional causality to determine whether aggression and political polarization have correlation ships. The proposed research project ultimately contributes to conceptual development for an internet community tracking system that provides the critical means for identifying early signs of social radicalization events that originate from social media. The proposed analysis helps in understanding the noise originating from social politics online using finance and machine processes.

3 Methodology

3.1 Data Preprocessing and Signal Isolation

To extract the needed data from the discrete interaction events suggested above, I implemented a pipeline to filter through the code execution logic.

3.1.1 Removing Ghost Strings and Fix Invalid Scores. The first layer in the pipeline, in the pipeline is to remove ghost-strings(low-entropy noises), and fix invalid scores.

Five rows (3 from democrats, 2 from republicans) were found to be invalid, and was fixed to valid numeric score of 0. Then, null pointers 'NaN', 'null', 'undefined' and single-character noise (e.g., "'") were deleted. A total of 163 rows, 79 from democrats, and 84 from republicans, were deleted. The sole use of "?" was kept because it has a semantic meaning in online communities as doubt or confusion.

The corpus was reduced by 163 rows, leaving 13,029,242 valid interactions.

3.1.2 Systemic Noise Differentiation. The second layer detects repeated posts that is possibly bot-created, ad, or systemic error messages and removes from the dataset.

The detection code filtered messages that are long and frequent messages that are more than 100 characters long and repeated more than 10 times. There were 201 unique ‘bot template’ messages found from Democrat data and 258 from Republicans, which consisted mostly of campaign ads and systemic messages.

The most common templates were:

- **Systemic Messages:** Automatic notifications regarding rule violations (e.g., *"Comment removed for breaking Rule 1"*) and subscription ping logs.
- **Campaign Mobilization:** Standardized calls to action, such as volunteer recruitment links and donation solicitations.
- **Ideological Spams:** High-frequency rhetorical blocks, including:
"The GOP is wholly and totally incapable of governing..." (repeated 145 times)
- **Hostile Automated Messaging:** Aggressive bot-driven alerts designed to harass users based on political affiliation.

Many other repeated spam messages were detected including links towards donation and private websites that lead to personal political statements.

A. Exclusion of External Propagation (Links/Ads)

All posts containing the `https://` schema were subjected to hard deletion. This resulted in the removal of 224,625 rows from the Democratic subset (4.5%) and 275,002 rows from the Republican subset (3.4%).

B. Semantic Normalization of Administrative Artifacts

Rather than deleting administrative messages (which would bias user activity metrics), we used Regex Pattern Matching to map them to neutral tokens. Rule violation texts were mapped to tokens such as `[removed_procedural_violation]` (Rule 1) `[removed_hostile_behavior]` (Rule 2) and more. Recurring scripts were similarly mapped to specific tokens (e.g., `[subscription_ping]`).

3.1.3 Identification of the Hyper Core.

A. Filtering ‘Super Users’

To make the dataset appropriate for the goal of this analysis, to track individual users in a time span, the code filters ‘Super Users’ with 500 or more posts to a separate file. This results in 2,495 unique users, excluding [deleted] users since they would be analyzed as a single clustered user and distract the analysis. The average member of the ‘Super Users’ had 951 posts, meaning that this specific online community system is dominated by 0.02% of users driving majority of the post volume.

B. Semantic Qualification and Filtering

While the gross volume was 2.37 million, not all interaction poses sufficient semantic density for high-dimensional encoding. It still consists of duplicate or short (50 or less characters) posts that are not sufficient to create a tendency or statement in the community. Filtering these validates a massive drop in the data volume, dropping the post number to 355,018. This removes about 142 posts per user.

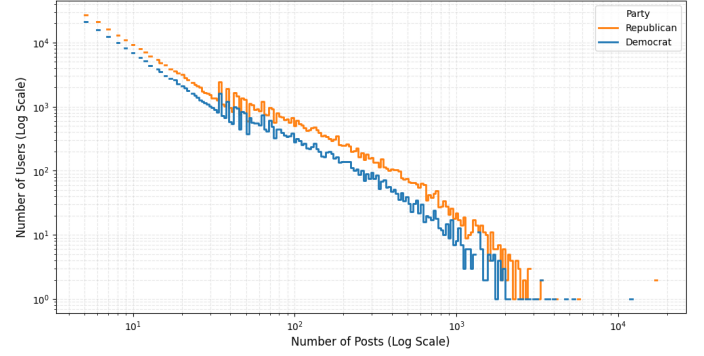


Figure 1: Post Volume per Users with 5+ Posts

C. Computational Stratification

Given the $O(N^2)$ complexity of the `all_mpnnet_base_v2` transformer, the data needs to be stratified to a sample. The pilot execution indicated an inference latency of 0.0331 seconds per post, and using a 3-hour window the sampling rate was calculated at 87.31%, meaning 309,956 posts.

Table 1: Data Filtering Pipeline

Stage	Action	Democrat	Republican	Total Rows
0. Date cut	Load Dataset	4,977,517	8,051,888	13,029,405
1. Structural	Ghost Removal	4,977,438	8,051,804	13,029,242
2. Spam	<code>https://</code> & Repeated	4,752,892	7,776,886	12,529,778
	Users w/ 500+			2,374,293
3. Hyper-core Scoping				355,018
	Sampling			309,956

Final Retention: 2.38% of original dataset

3.2 Semantic Feature Engineering

To compute political communication through a rather radical online channel, using traditional lexicon-based approaches like VADER or TextBlob was insufficient. These methods rely on keyword dictionaries, which makes it hard for them to detect sarcasm and contextual meanings. Instead, using a Sentence-BERT(SBERT) framework which uses attention mechanism is more appropriate for this task.

We utilized the pre-trained `all_mpnnet_base_v2` model. This architecture maps variable-length text sequences into a 768-dimensional dense vector space (R768), such that semantic similarity is preserved as a geometric distance.

3.2.1 Aggressiveness Metric. To quantify the degree of hostility in user comments, I developed and continuous aggressiveness metric. This metric leverages domain-adaptive semantic projection, mapping user texts to a definition of aggression derived from the community’s vulgar.

A. Seed Generation

By employing a ‘CountVectorizer’ to generate the vocabulary set V , consisting of the top 1,000 most frequent unigrams, the code successfully filtered standard english stop words in the dataset. This constraint makes the semantic space to be constructed out of the terms actively used in the population.

The minimal initial seeds were manually defined as ["idiot", "stupid", "liar", "shame"] and ["civil", "agree", "understand"] each for aggressive and calm.

Then I generated vector embeddings for these words using the SBERT model, then applied centroid-based expansion to generate two opposing word sets and measured the cosine similarity to find 40 nearest words for each pole. These are some of the words that I actually used to detect the anchor sentence: seed_aggressive = ['stupid', 'dumb', 'bullshit', ...], seed_calm = ['understand', 'agree', 'civil', ...]

B. Anchor Sentence Generation

Using the centroid among the seed words created above, the concept of hostility was created. Then by measuring the distance from this centroid to all the user comments by cosine similarity, I was able to score them in a scale from 0 to 1. Then by grabbing the top 100 most aggressive comments, and filtering the length(for diversity and sufficiency), K-Means Clustering distinguished them to 10 different groups. One sentence from each group was selected as the anchor sentence. This process was to keep diversity, and to avoid sentences like 'You are stupid' to be repeated among the anchor sentences.

These were the final anchor sentences created:

Table 2: Semantic Anchors Sentences

Category	Exemplar Anchor Sentences
Aggressive	"I'm mad I ever supported this piece of shit..."
	"A pure maga moron spewing hate, disinformation and lies."
	"FAKE NUMBERS. FAKE NEWS!!!!11!!!!11!!!! /S"
	"What a fucking insane thing to say. Hard to grasp."
	"Probably more of a Russian troll."
Calm	"He lied on voter forms multiple times."
	"I agree. Lets take a poll on the question."
	"Not sure you made a point of discussion at all."
	"You may not agree with it but that's just part of the process."
	"We must support each other publicly and privately."
	"I'm not a fan of reason but they are pretty consistent with their ideology"
	"This is not going anywhere useful."

C. Metirc Construction via Weak Supervision

To translate the semantic insights to a continuous numerical metric, I employed weak supervision strategy. This process I created changes geometric vector projection into a probabilistic model that indicates the hostility used in user comments without human annotation.

The heuristic axis in the vector space was made by the centroid of each anchor sentence group. Every comment in the dataset was projected to this axis using cosine similarity, most aggressive data projecting to 1 and calmest data projecting to 0.

Since this raw scores might not be well distributed enough, I assumed that the top 10% and bottom 10% of the scores were

definitely aggressive and calm, and labeled their scores as 1 and 0.

I went ahead and trained the Logistic Regression classifier on this weakly labeled data. At this stage, the aim was calibration, not discovery. We can train the classifier on the endpoints to tune the sigmoid function to our data. This enables the algorithm to find the slope of the curve from civil to aggressive speech and to predict the chance score of the ambiguous cases, which fall in the centre of the distribution. The model obtained an accuracy of 100% on the test set of the weakly labeled data. This accuracy is structural, rather than a consequence of overfitting. Since the labels were created by partitioning based on a linear function of cosine similarity, stripping away the middle 80%. By definition, the two classes, Top Decile and Bottom Decile, are linearly separable.

The Logistic Regression Model was able to accurately identify this hyperplane. The usefulness of this process will be found in its ability to map this unlabeled 80% data onto this developed sigmoid function, grading these grey area comments on the comments that have very distinct characteristics from these obvious extremes.

3.3 Polarization Metric

In order to assess the level of political bias of each comment, I have created a supervised machine learning classifier that can identify Democratic and Republican ideological biases. This is in contrast to keyword matching algorithms that tend to confuse the subject of discussion and its corresponding bias. This is done through the utilization of contextual embeddings that help in understanding the semantic sentiment in texts.

A. Creating Training data

I constructed a labeled training dataset by isolating comments from definitely polarized subreddits, which I know it's political sides they're on. Subreddits such as r/politics and r/news are where both parties are active, so I chose subreddits below that are known to be packed with each party supporters:

```
dem_subs = ['VoteDEM', 'democrats', 'WhitePeopleTwitter',
'PoliticalHumor', 'Political_Revolution', 'SandersFor
President', 'EnoughTrumpSpam', 'progressive']
```

```
rep_subs = ['Conservative', 'trump', 'conservatives',
'Republican', 'ShitPoliticsSays', 'AskThe_Donald',
'ConservativesOnly']
```

To guarantee the posts are arguably agreed within each subreddit, I chose data those have scores(their net upvote) higher than 10. This yielded a high confidence training set of 27,620 comments, 18,419 democratic samples and 9,201 republican samples. They were each temporarily labeled 0 and 1 to create a heuristic axis.

B. Model Architecture and Training

Using a logistic regression classifier, the code predicted the ideological alignment of a comment based on its semantic

vector. It utilized the BERT based vectors created above. I also applied L2 regularization ($C=0.5$) to prevent over fitting. The trained model was then applied to the entire dataset to generate a continuous polarization score, representing the probability of a comment belonging to the Republican class.

3.4 User Clustering and Identification

After all the superuser's comments were labeled with their probability towards aggressiveness and polarization, I then applied unsupervised machine learning to segment the users into distinct archetypes defined their ideology, verbal features, and community impact. This enables the identification and analysis of the active users.

To all the users in the data set, I computed their mean scores for polarization, aggression, and engagement, which was represented in the field of net upvote score. I implied K-Means Clustering to segment the users within the three-dimensional feature space of the three means I calculated. The features were applied Robust Scaling to minimize the distorting effect of extreme outliers. I selected $k=4$ as the target number of clusters, and the algorithm was run with $\text{random_state}=42$ and $\text{n_init}=10$ to ensure reproducibility and convergence.

3.5 Causal Dynamics and Temporal Inference

To analyze whether aggression drives partisan identity or partisan identity drive aggression, I moved towards the long term time series analysis. Using Granger Causality test, the code will determine if there is a signal between the two metrics over time.

3.5.1 Time Series Aggregation and Stationarity. I turned the interaction vectors into discrete daily time steps ($t = 724$) to capture the trend of the two fields, Aggression score (A_t) and Polarization score (P_t). Before the causality test, I assessed the stationarity of the signals using the Augmented Dickey-Fuller (ADF) test. The ADF test posits a null hypothesis that the series has a non-stationary unit root. If the resulting p-value is below the threshold of $\alpha = 0.05$, it means the data is safe for modeling.

The polarization scores had the p-value of 0.000, meaning it rejects the null hypothesis with high confidence and was stable with stationary process. The aggression scores yielded a p-value of 0.0162, also yielding stationary process. If either of the results were non-stationary, I'd have to apply first-order differencing to both metrics.

3.5.2 Granger Causality Tests. To account for behavioral divergency, I stratified the analysis into two cohorts; the system-wide population and high-risk agitators. I defined the Granger Causality model to test if changes in one metric provide significant information about future changes in the other metric across the lag of days.

4 Results

4.1 Semantic Scores

The SBERT based feature engineering successfully projected 309,956 posts made by the core 2,495 population onto the calibrated aggression and polarization axes.

Table 3: Model Scoring Examples

Pol.	Agg.	Post Text
0.7200	0.4740	Gaetz didn't fail because he was going to help Trump be dictator, he failed because people in the senate hated his guts.
0.0059	0.8305	Don't confuse us with r/politics and r/Biden filled with right-wing extremists and fascist bots like yourself.

4.1.1 Aggression Scores. The aggression scores show high frequency near 1.0 and 0.0. This might be a problem with the model not distinguishing the texts evenly, but certainly not an error because it was found that no posts were actually labeled 1.0 or 0.0. The fact that the graph is extremely polarized suggests that the posts in political online communities in reddit have a very binary character. The Aggression score trend over time is very clearly shown that

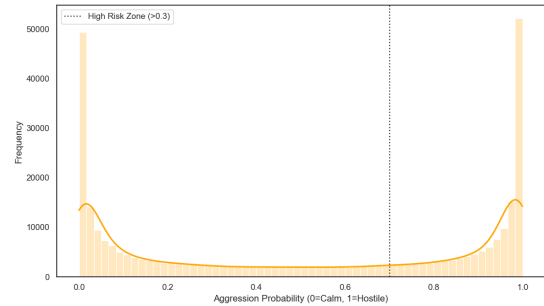


Figure 2: Distribution of Aggression Scores

it gradually increases over time until the October of 2024, just before the presidential election. Then after the election, aggression dropped quickly (with some data missing too) and experience a huge bump in August of 2025. But compared to the late 2023 graph, the overall aggression is highly increased, with the 2024 election period (December 2023 to October 2024) being the most aggressive of all.

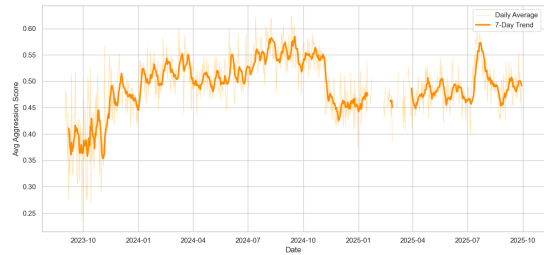


Figure 3: Trend of Aggression Scores

4.1.2 Polarization Scores. The distribution of the polarization scores show the ideological atmosphere in the overall reddit community. As the graph suggests, it is a bit more right-wing focused, while keeping moderate overall. This is because the training was made within the data itself, so comparison with outer online communities weren't made.

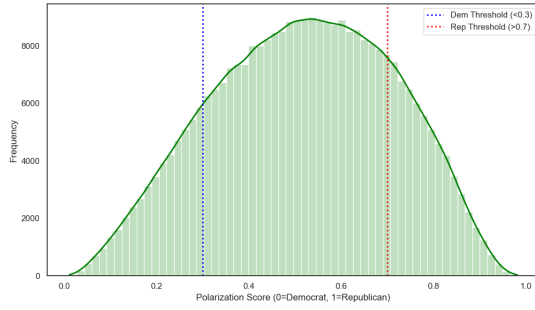


Figure 4: Distribution of Polarization Scores

The polarization trend moved toward Republicans overtime, which most likely means more Republicans are flowing in considering the increasing post volume and users. The movement towards the right wing happened as the 2024 election period began, and the hump in August 2025 was also detected.

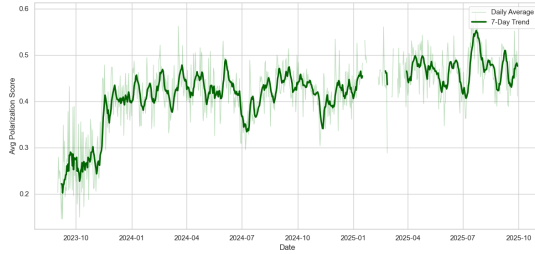


Figure 5: Trend of Polarization Scores

While Fig 5. suggests the tendency of online reddit communities moving towards the right wing, the standard deviation graph shows how the users are polarized towards the extreme ends. It is clear that extremists increased around December 2023, and stayed mostly that way.

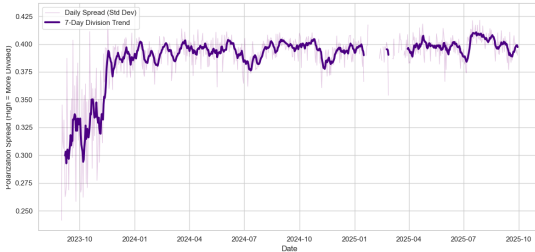


Figure 6: Trend of Polarization Standard Deviation

By the overlapping graph of both scores, we can tell the polarization and aggression scores follow the similar trend, meaning right-wing biased posts are what derive most of the aggression. However from July 2024 to November 2024, it is shown that the increase of aggression doesn't quite follow the polarization score. This means that during this time, the Democratic users also engaged a lot in aggression. And from November 2024 to July 2025, the Republican biased posts were less aggressive.

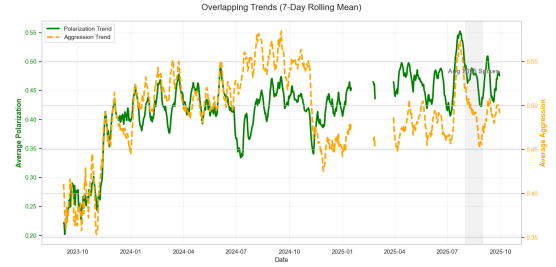


Figure 7: Trend of the two Scores

4.2 Structural Segmentation

The K-Means clustering on the three dimensional feature space of polarization, aggression, and engagement, segmented the population into four different groups.

Table 4: User Group Metrics: Aggression, Polarization, and Engagement

Group Number	Avg. Aggression	Avg. Polarization	Avg. Net Engagement	User Count
1	0.6368	0.5680	329,671	818
2	0.5697	0.5134	1,008,466	206
3	0.4916	0.3872	182,720	1,012
4	0.2907	0.2398	52,398	459

Note: Avg. Net Engagement is calculated per user cluster.

I labeled each as Partisan Front-Liners, Systemic Anchors, Passive Mainstream, and Civil Minority according to their features. The Systemic Anchors(Yellow) are the most minority with the highest average net engagement, meaning they are the very few 'influencers' of the community. You can tell they are mostly right-wing aggressors, with net upvotes easily up to 1 million, about 19 times more than the Civil Minority.

While the Partisan Front-Liners(red) are another majority, with high aggression and polarization score than any other groups. They are heavily centered in the right-wing, with the population of 32%.

The Passive Mainstream(green) are the biggest population in moderate left wing. Civil Minority(blue) are the ones with the lowest aggression and polarization scores, and the lowest net engagement. They are the heavily left weighted segment that are calm and less influencing.

This is not the overall trend of all the users, since the sample is in fact within the 'superusers' who have 500+ posts. However, this shows the leading and most engaging people of the online community, and what kind of users get to influence the overall atmosphere of this online debate.

As you can tell from the trend of two scores by each segment is very different. The four clusters tend to stay distinct with each other in the over all period, while slightly following the overall trend like increasing aggression in August 2025 and rapidly decreasing in November 2024.

To compare the most active two groups, the Partisan Front-Liners and Systemic Anchors, I aligned their trend with the overall average

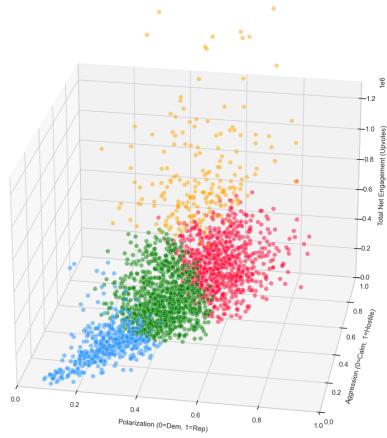


Figure 8: User Segmentation in 3D Feature Space

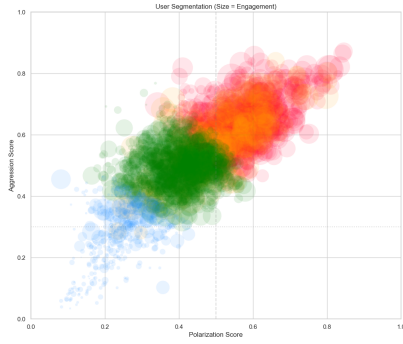


Figure 9: User Segmentation in 2D Feature Space

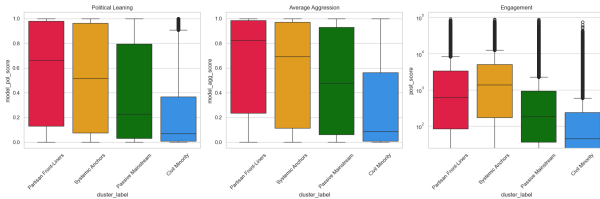


Figure 10: Bar Chart of User Groups

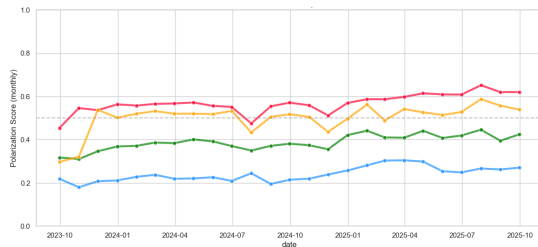


Figure 11: Trend line graph of Polarization Scores by User Groups

trend. They follow a very similar sequence, while keeping higher scores than the overall.

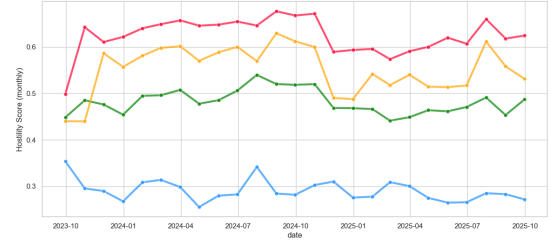


Figure 12: Trend line graph of Aggression Scores by User Groups

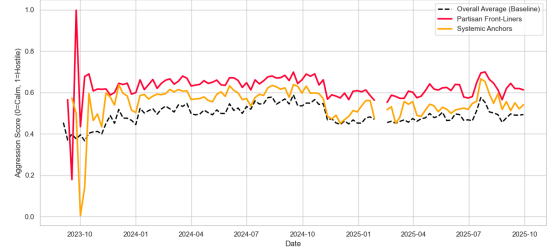


Figure 13: Aggression Score comparison between High-Risk Core and General Public

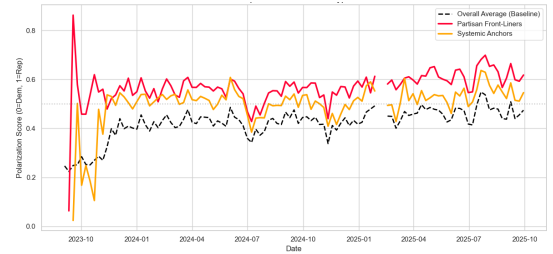


Figure 14: Polarization Score comparison between High-Risk Core and General Public

4.3 Causal Inference

To determine whether ideology drives anger or anger drives ideology, I used the granger causality test by analyzing daily data points. I looked for instances where the history of Variable X can statistically predict the future of Variable Y in various data clusters, and to determine the lagging time.

4.3.1 General Public. Amongst the superusers in political subreddits, the results showed that aggression significantly granger causes polarization at $Lag2(p = 0.029)$ and $Lag3(p = 0.012)$. This means that the general public amongst the superusers experience a lagged radicalization, where hostile interaction causes a shift of partisan leaning in 2-3 days. For the general users, the emotional environment causes polarization. But the correlation is a weak negative of -0.0061 , meaning this timing is predictable while the direction is not quite clear, moving inconsistently.

4.3.2 High-Risk Core. For the two most influential cluster of Systemic Anchors and Partisan Front-Liners, the group of 1,024 users, the polarization shifts granger cause aggression in $Lag2(p < 0.0001)$ and $Lag3(p < 0.0001)$. This cluster experience aggression after

their political bias in the lag of 2-3 days, while their correlation is a moderate negative of -0.3939 .

This means that these users show high relationship of polarization and aggression, where if they slide to democratic topics and statements, their anger increases. They tend to attack the democratic viewpoint, while feeling comfortable with republican viewpoints.

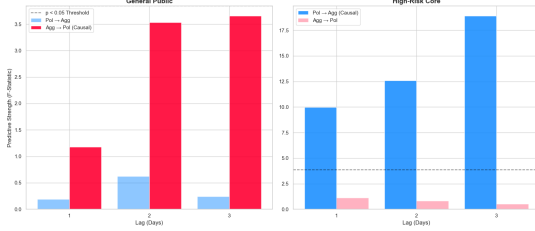


Figure 15: Granger Causality Results Summary

5 Discussion

The findings of the present study suggest that, in the Hyper-Core subreddit, radicalization and hostility are predictable systemic outcomes of the incentive structure on the site and not random behavioral glitches.

5.1 Binary Volatility and the Election Cycle

The temporal analysis proves that hostility and polarization in the online community are sensitive to the exogenous events of the political calendar. The steady increase in aggression up to November 2024 presidential election, and the scores being maintained suggest that not only did the community become active and aggressive amongst this season, the aggression and polarization stayed this way through the presidency of Donald Trump. Although the post election cooling was observed, the aggression stayed higher than it was before. This indicates a “ratchet effect”, where each election cycle raises the community’s baseline of both aggression and polarization. This can also suggest that the users’ inflow during this season is not a temporary phenomenon, but the users stay active in the community after this event.

The semantic score during the time period after the election suggests something other. While during the election season, polarization stayed in the similar level as aggression, meaning it was the Republican opinions that derived aggression in the community. However, after the election the polarization score didn’t quite follow up the aggression score, meaning the Democratic aggression increased, especially spiking in July to September of 2025. This suggests that many democratic users showed aggression after the Trump administration.

5.2 Analysis of the Causal Inversion

The most significant factual finding is the symmetrical inversion of causality between the general public and the high-risk core group. This suggests that the radicalization of the online political community is not a mere single process, but two distinct psychological phenomena that are highly correlated with another.

For the majority of the superusers, the *Lag3* result for aggression to polarization proves that hostility is an essential factor for the community to lean on the right wing. This suggests that the subreddits serve as a heated online environment that accords the users’ emotional baseline. The 3-day lag suggests that this is the time span needed for the majority of the users to turn aggressive interaction to the shift towards the right political pole.

However, the high-risk core group reacts differently toward the online environment. They turn right-wing polarization towards aggressiveness with the lag of 2 days. This suggests that these users are already polarized to a specific partisan pole, and when ever they start leaning to the left wing they react in an aggressive way to the opposing argument. By the fact that they tend to have a strong relationship with the majority opinion, this type of aggressive reaction is what influences the whole community.

5.3 Structural Risk Assessment

In order to confirm whether the displayed aggression was an intrinsic characteristic of the social architecture within this community or not, I applied an XGBoost Predictive Model augmented by SMOTE (Synthetic Minority Over-sampling Technique). I aimed to find out if being high-risk (i.e., having an average aggression score >0.6) can be assessed based on purely metadata-driven user characteristics, namely ideological orientation, engagement level, as well as number of messages posted, with no actual textual analysis necessary on their messages.

5.3.1 Predictive Determinism and the Clarity Gap. The model achieved an overall accuracy of 62%. While the model is not a perfect classifier, its performance reveals a significant metadata fingerprint for radicalization. It achieved a recall of 0.54 for high-risk users, meaning it successfully identified more than half of the community’s agitators based purely on their structural footprint.

The average risk probability calculated by the model had a clarity gap of 14.8% (0.575 for agitators and 0.427 for general users). This is a key result because it verifies the hypothesis stated in the introduction, that the system itself has a constant gravitational pull toward aggression.

Before even checking the actual text of the user post, the user’s position in the network like their politics and clout, the person can be statistically predisposed toward hostility. Yet the existing overlap between the groups signifies a human agency factor, where approximately 40% of the users’ demographic profile of agitators use civil discourse.

Table 5: Structural Risk Assessment Model Performance (XGBoost + SMOTE)

Metric Class	Precision	Recall	F1-Score	Support
0 (Safe Users)	0.78	0.66	0.71	532
1 (High-Risk Agitators)	0.39	0.54	0.45	217
Accuracy			0.62	749
Macro Average	0.58	0.60	0.58	749
Weighted Average	0.67	0.62	0.64	749

5.3.2 The Determinants of Risk. By analyzing the gain, which is the contribution of each feature to the model's accuracy, we identified the primary drivers of community risk.

Average net engagement, which is the upvotes of each post, emerged as the most powerful predictor. This confirms the subreddit's incentive structure functions as a toxicity accelerator. Users who receive high validation and attention in the community are mathematically more likely to engage in high-risk behaviors.

Polarization mean is also a primary driver to community risk. This means that user's movement toward ideological extremes leads to escalating risk.

Another factor was the consistency of a user's ideology and engagement, which was represented in the standard deviation of the two fields. This represents the volatility, which by the results suggests that high-risk users often exhibit inconsistent spiky behaviors with sudden bursts of extreme partisan rhetoric or viral hostility.

5.3.3 Implications of the SMOTE Upgrade. The role of SMOTE in rebalancing the training set was crucial in revealing these relationships. Without rebalancing, the system was biased towards ignoring the minority agitator class in favor of the safe majority class. By artificially populating the minority class, we challenged the system to identify the distinctive characteristics of the "Partisan Front-Liner" over the "Passive Mainstream" user. This is because the model has now achieved 0.39 Precision in relation to high-risk users, which clearly indicates that although metadata is a potent predictor, the community still has "Potential Agitators" who possess the extreme ideology and considerable power but have stayed short of toxic levels.

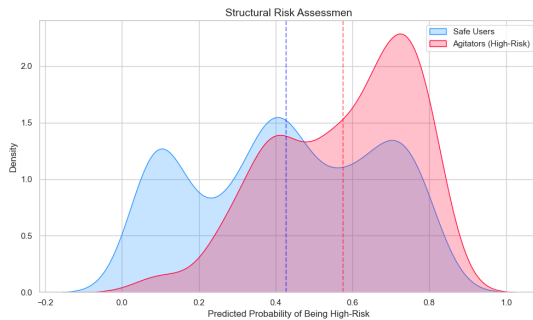


Figure 16: Structural Risk Assessment

6 Conclusion

This paper finds that radicalized behaviors in online political groups can be predicted, including their activeness, by the inherent design of the site rather than being some random occurrences. This paper deconstructs the hyper-core region found on the subreddit website to show what the logic may be behind the radicalized behaviors found on the online environment.

The main results are: symmetrical inversion of causal relationships between the public and the high-risk group's influential mainstream, while the public uses the affective route to radicalization, hostility is the gateway to emotions regarding the ideology that

attunes to it within three days; aggression is a tool of the ideology wielded by the high-risk elite within the community as a result of its imbedded motive of aggressive rhetorical action.

Also, the proper workings of the XGBoost prediction model validate the implication that there is a metadata fingerprint of radicalization. Given the fact that one can identify high-risk individuals with 62% accuracy based only on the factors of community engagement and polarization, this paper shows without any shadow of a doubt that toxicity is indeed a measurable outcome of the incentivization mechanism for the community.

Thus, specifically, the fact that the 14.8% clarity gap of probability of risk remained exposed by this study indicates, to a certain extent, that the concept of individual free will is no longer applicable, in the sense that there is a constantly present gravitational pull of mass culture, which eliminates such will.

Ultimately, the final objective of this research will be used as a foundation for conceptualizing a model for online communities being tracked online. With these structural elements of tip in place, a proactive understanding of online communities beyond text-based moderation will be ensured as a means for online sociologists to further understand the toxicity premium of viral hostility as it drives the process of public radicalization online.

7 Reference

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